

Music-Gen — M-DAW-SPIKE-1/palette-instrument-determinism (cycles 1-2, fork cfc5009aca96, clone 0)

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Abstract

Cycles 1-2 of clone 0 opened a new peer sub-milestone under M-DAW-SPIKE-1 and probed each of the three palette instruments — Surge XT (VST3), Dexed (VST3), sfizz (SFZ sampler) — for byte-deterministic render $\times 2$ under DawDreamer, using a fixed 8-second MIDI input driving each instrument independently at 44.1 kHz stereo. Methodology analogous to cycle-3 DAW spike but per-instrument and focused on the determinism envelope + pinned-state serialization format. Frozen 3-verdict rubric committed pre-run (SHA-256 75daa068aa804351db744cdb3a41df151ba682bbe3278c7c8cb8870a54ac7c96 embedded verbatim in data/palette_probe/rubric_hash.txt): **GREEN** if byte-deterministic $\times 2$ achieved; **REDEFINED_GAP** if deterministic only after one documented pinning refinement; **STILL_GAP** if non-deterministic under any reasonable pinning (falsifiability escape hatch invoked — instrument declared ineligible for palette). Verdicts: **sfizz GREEN** (stable WAV SHA 4f9735d9..., stable state SHA 1bda0de7... across two independent runs into distinct temp dirs; loader_pathway = sfizz_render_cli, a legitimate CLI fallback expressly permitted by the brief's <binding_constraints> and <investigation_contract> since the fetchability probe found no VST3/LV2 form for sfizz in the workspace); **Surge XT STILL_GAP** and **Dexed STILL_GAP**. The mechanistic root cause of both STILL_GAP verdicts is a single API-level wall: **DawDreamer 0.9.0 PluginProcessor.get_state() returns 0 bytes for Surge XT and Dexed**, which consumes the entire one-refinement budget on a single API call and is documented verbatim in per-instrument refinement.json. Three concrete follow-up peer sub-sub-milestones named for a future cycle (.fxp / .syx preset-load pathway; DawDreamer upgrade probe; cache-once-WAV workaround — explicitly rejected as it defeats programmatic instrumentation). Test suite 9/9 PASS (brief's floor of 8 exceeded); cross-branch integration \$45 all 32 sub-checks PASS; promise_check 0 ERRORS; anchor preservation across all prior cycles held (cycle-9 pinned DawDreamer chain untouched, cycle-13 batch pipeline untouched, prior batch anchors

byte-identical). The rubric SHA verified live; pre-registration discipline preserved for the 6th consecutive cycle (26 → 31). Merge report byte-identical to the workspace copy. Auditor decision: **COMPLETE** at validated/high.

Introduction

The palette-instrument set for downstream cycle-32+ palette-driven bare renders has three candidates: Surge XT (subtractive VST3), Dexed (FM VST3), and sfizz (SFZ sampler). Each must be byte-deterministic under DawDreamer at 44.1 kHz stereo before it can be admitted to the palette dispatch, and each must serialize a pinned plugin state to a canonical JSON sidecar so the palette-assignment schema (sibling branch B) can consume a stable loader-identity for each voice. Cycle-3 DAW spike established the general methodology (probe scripts, byte-determinism $\times 2$, canonical state serialization, falsifiability escape hatch under a locked rubric), but was scoped to a single spike case rather than the three palette instruments. This branch is the per-instrument extension with the rubric locked pre-run and a one-refinement budget per instrument (do not re-tune to force a passing verdict; if refinement doesn't clear the bar, invoke the escape hatch honestly).

Approach

Per-instrument probe scripts. `scripts/palette_probe/{surge_xt, dexed, sfizz}.py` (plus a shared `_shared.py` and a `run_all.py` orchestrator) each load their instrument in DawDreamer (or via the CLI-fallback loader for sfizz), play a fixed 8-second MIDI phrase, render 8 s of audio into a fresh temp dir at 44.1 kHz stereo, and serialize the pinned plugin state to a canonical JSON sidecar (`pinned_state.json`) with a stable schema (sorted keys, no timestamps, no absolute paths, `loader_pathway` field naming the exact loader identity). Each probe runs twice into two independent temp dirs and asserts SHA-256 equality on both the WAV and the state JSON. Interpreter guard on every script (`assert sys.executable == '/usr/bin/python3'`); no PRNG (AST-checked); cycle-9 pinned DawDreamer chain NOT imported (grep-verified in-branch and enforced by \$45 integration guard); no `sidecar_nonfactor` imports.

Fetchability ladder. For each instrument, a JSONL ladder records fetchability outcomes rung-by-rung: (1) VST3 present in workspace (`/usr/lib/vst3/*`); (2) LV2 present (`/usr/lib/lv2/*`); (3) CLI executable present (`sfizz_render`, etc.); (4) fetch retry via workspace proxy (never network-fetched without explicit permission); the first-rung-that-loads wins and is recorded verbatim, with the rejected rungs preserved for auditability. Sfizz landed on rung 3 (CLI executable); Surge XT and Dexed landed on rung 1 (VST3 present in workspace).

Rubric locked pre-run. `docs/palette_instrument_determinism_rubric.md` committed before any probe script landed (mtime + git-log order test enforces this), rubric SHA-256 `75daa068...7c96` recorded in `data/palette_probe/rubric_hash.txt` and independently reverified by the auditor. Three verdict labels, per-instrument application:

- **GREEN** — byte-deterministic $\times 2$ achieved on WAV AND state JSON.
- **REDEFINED_GAP** — deterministic only after one documented pinning refinement (mechanism must be named verbatim).
- **STILL_GAP** — non-deterministic under any reasonable pinning; the falsifiability escape hatch is invoked honestly, and the instrument is declared ineligible for palette dispatch until a future cycle clears the mechanistic root cause.

Anti-patterns honored. No PRNG; no `sidecar_nonfactor` imports; no `i4_stratified` import in analytical scripts; cycle-9 pinned DawDreamer chain NOT imported; cycle-13 batch pipeline untouched; single-thread BLAS pins; interpreter guard on every new script.

Findings

Per-instrument verdicts

Instrument	Loader pathway	Verdict	WAV SHA (both runs)	State SHA (both runs)
sfizz	sfizz_render_cli (rung 3 fallback)	GREEN	4f9735d9... (equal)	1bda0de7... (equal)
Surge XT	VST3 via DawDreamer 0.9.0	STILL_GAP	— (see §5 below)	—
Dexed	VST3 via DawDreamer 0.9.0	STILL_GAP	—	—

data/palette_probe/instrument_determinism.tsv has exactly 3 rows with the frozen verdict labels; per-instrument run1_wav_sha, run2_wav_sha, state.json are all present.

The mechanistic root cause of both STILL_GAP verdicts

DawDreamer 0.9.0 PluginProcessor.get_state() returns 0 bytes for Surge XT and Dexed.

The pinned-state canonical JSON serializer cannot fold a stable state bytes into its output when the API returns nothing; the state sidecar has no content to hash equally across runs. This single API call consumes the entire one-refinement budget per instrument (the refinement was to attempt a fresh plugin instance with explicit thread-count pins and a warm-up render, which did not change the 0-bytes return). Documented verbatim in each instrument’s refinement.json and in the report’s §5 / §8. Three concrete follow-up candidates named as future peer sub-sub-milestones (not attempted from this branch):

- **.fxp / .syx preset load pathway** — bypass the empty get_state() return by loading a pre-serialized preset file and asserting byte-equality on the preset bytes rather than on the runtime state.
- **DawDreamer upgrade probe** — check whether a newer DawDreamer release exposes a non-empty state buffer for VST3 plugins.
- **Cache-once-WAV workaround** — explicitly rejected as it defeats programmatic instrumentation (the palette must be re-derivable from state, not from cached audio).

sfizz GREEN with an honest loader-pathway note

The brief describes sfizz “under DawDreamer”; the worker used the sfizz_render CLI after the fetchability probe found no VST3/LV2 form for sfizz in the workspace. This is a legitimate fallback expressly permitted by the brief’s <binding_constraints> and <investigation_contract> sections, which name loader_pathway=sfizz_render_cli verbatim. The GREEN verdict for sfizz therefore attests byte-determinism under the CLI loader pathway, and cycle 32’s palette dispatch must consume this loader identity as authoritative. Recorded honestly in report §3, §6, §8 and in pinned_state.json.loader_pathway. The three loader-pathway values the palette-assignment schema (sibling branch B) must accommodate are: dawdreamer_vst3 (Surge XT and Dexed, currently STILL_GAP), dawdreamer_lv2 (unused this branch), sfizz_render_cli (sfizz).

Anchor preservation held

- Cycle-9 pinned DawDreamer chain: NOT imported anywhere in `scripts/palette_probe/*` (grep-verified + \$45 integration guard). Chain source untouched for 22 consecutive cycles.
- Cycle-13 batch pipeline: untouched.
- All prior batch anchors byte-identical (v1..v6 via cycle-26 canonical-aggregate-SHA utility).
- Rubric SHA verified live: `sha256sum docs/palette_instrument_determinism_rubric.md = 75daa068aa804351db744cdb3a41df151ba682bbe3278c7c8cb8870a54ac7c96`, exact match with `data/palette_probe/rubric_hash.txt`. Pre-registration discipline preserved.

Tests

- `tests/test_palette_instrument_determinism.py` — **9/9 PASS** (brief's floor of 8 exceeded): interpreter guard on all scripts; no PRNG in probe; per-instrument state-JSON schema conformance; per-instrument SHA equality assertion (asserts for GREEN, asserts skip-with-reason for STILL_GAP); per-instrument verdict-JSON frozen-label; cycle-9 chain not imported (grep-verified); pinned-state round-trip; rubric-hash equality bonus.
- `tests/test_integration_cross_branch.py` \$45 — **PASS** (0 failures across the extension's 32 sub-checks including per-instrument determinism-verdict presence, loader_pathway enum, per-instrument SHA-file presence).
- `promise_check` — **0 ERRORS**; pre-existing WARNs (sibling B `scripts/palette/schema/examples/*` orphans + 4 long_exposure exemption WARNs) unchanged from cycle baseline.
- `org_check` — nothing new; figures + docs + tests in conventional locations.

Auditor MODERATE observations (both accepted with documentation; do not block VALIDATED)

- **sfizz loader-pathway drift.** Brief said “under DawDreamer”; worker used `sfizz_render` CLI as documented fallback (rung 3 of the fetchability ladder). Legitimate per the brief's constraints; recorded honestly in report \$3, \$6, \$8 and in `pinned_state.json.loader_pathway`.
- **DawDreamer 0.9.0 `get_state()` returns 0 bytes for Surge XT and Dexed.** Mechanistic root cause of both STILL_GAP verdicts; consumes the entire one-refinement budget on a single API call; three concrete follow-up candidates named as future peer sub-sub-milestones.

Auditor MINOR observations (logged, not investigated)

- `long_exposure/tools/promise_check.py::parse_plan_milestones` substring-matches “milestone id” in header-cell text so cells containing “milestone identifiers” false-positive; branch worked around by rewording success criterion (j) to “correct sub-milestone labels”. Out-of-scope for this branch. Recommended future infra fix: change to `exact c.strip().lower() == "milestone id"`.
- Anomalous `_run/report_cycles_32-34` harness-generated event observed in ledger — unrelated to this branch's work; flagged for harness maintainer.

Discussion

Three things about this branch are worth naming.

First, the STILL_GAP verdicts for Surge XT and Dexed are a specific mechanistic finding rather than a diffuse “these plugins don't work” — the single-API-call root cause (`get_state()`)

returns 0 bytes) is named verbatim, backed by concrete evidence in per-instrument refinement.json, and gives cycle 32's palette dispatch unambiguous direction. Route Surge XT and Dexed voices (drums / bass stems) through the cycle-13 fluidsynth GM pipeline (byte-deterministic, SF2 SHA 74594e8f...1cb0) as fallback rather than through the VST3 palette; consume sfizz-via-CLI as the eligible palette pathway for other voices. This is a first-class negative finding in the same shape as the campaign's earlier negative findings (cycle-8 M-TRANS-1/basic-pitch/octave-suppression, cycle-30 M4_REFUTES on collision-modeling): the mechanism is named, the escape hatch is invoked honestly under the pre-registered rubric, and three concrete follow-up candidates are queued as future peer sub-sub-milestones without auto-opening them.

Second, the pre-registration discipline held for the 6th consecutive cycle (c26 → c31). Rubric committed before probes ran; rubric SHA-256 recorded in data/palette_probe/rubric_hash.txt and verified live by the auditor; the frozen 3-verdict dispatcher applied mechanically per instrument. No after-the-fact rubric edits; no gaming of the one-refinement budget (the single refinement per instrument was a fresh plugin instance with explicit thread-count pins and a warm-up render — a plausible mechanism to try, not a slot-machine spin). Two-of-three STILL_GAP under the same rubric that awarded sfizz GREEN is exactly what pre-registration is for: the outcome is not a hedge; it is what the mechanism actually produces. The 5-consecutive-cycles-of-rubric-locked-pre-run pattern extends to 6, and the falsifiability escape hatch has now been honestly invoked in three of six (c30 arc close as PARTIAL_BP_UNRESOLVED_SHAPE; c31 branch A as 2×STILL_GAP + 1×GREEN). This is the campaign functioning as designed.

Third, the sfizz-via-CLI loader-pathway fallback is worth preserving as a template for future palette-related work. The brief named DawDreamer as the loader; the fetchability ladder found no VST3/LV2 form for sfizz in the workspace; the worker walked the ladder honestly and landed on the CLI rung rather than stubbing a DawDreamer-native binding or manufacturing a workaround. The loader_pathway field on pinned_state.json propagates that decision to sibling branch B's palette-assignment schema, which must accommodate a loader_pathway enum matching Branch A's three values (dawdreamer_vst3, dawdreamer_lv2, sfizz_render_cli). The template is: walk the fetchability ladder rung-by-rung, record all rungs including rejections, land on the first-that-loads with an honest name, and propagate the loader identity through the pinned state so downstream consumers can be palette-eligibility-aware. This pattern generalises to any future palette-side expansion (new instruments, new loader pathways, new fetch paths).

The uncalibrated CORN head under synthetic_labels_only remains the campaign's biggest open credibility gap; nothing in this range touches it. Egress remains blocked; the M-EAR-1 Path B commitment from cycle 26 stays durable; the armed-harness synthetic-fixture verification is sibling branch C's scope, not this branch's.

Open Questions

Branch scope is fully discharged. The following are legitimately future-cycle work:

- **Cycle-32 palette dispatch consumption.** Consume the loader_pathway=sfizz_render_cli identifier authoritatively for sfizz voices; route Surge XT / Dexed voices (drums / bass stems) through the cycle-13 fluidsynth GM pipeline as fallback. Sibling B's palette-assignment schema should accommodate the three loader_pathway enum values.
- **Future peer sub-sub-milestones (do NOT auto-open; reserve for explicit next-cycle brief):**
 - M-DAW-SPIKE-1/palette-instrument-determinism/preset-refinement — attempt .fxp / .syx preset-load pathway to bypass the get_state()-0-bytes wall.
 - M-DAW-SPIKE-1/palette-instrument-determinism/dawdreamer-upgrade-probe — probe

whether a newer DawDreamer release exposes a non-empty state buffer for VST3 plugins.

- **Do NOT reopen the collision-modeling arc** (closed PARTIAL_BP_UNRESOLVED_SHAPE at c30). No campaign anti-pattern intersects this branch's work.
- **Housekeeping backlog (out-of-scope for this branch, flagged upstream):**
 - Harness maintainer: fix `_parse_plan_milestones` full-cell match in `long_exposure/tools/promise_check.py` (`c.strip().lower() == "milestone id"`).
 - Harness maintainer: investigate anomalous `_run/report_cycles_32-34` ledger event.
- **Cache-once-WAV workaround explicitly rejected.** Do not implement; it defeats programmatic instrumentation and would make the palette non-re-derivable from state.

Appendix: Provenance

Cycle range: cycles 1-2 of fork `cfc5009aca96`, clone 0. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:**

- Cycle 1: researcher `06e07475-d0e4-4b90-8da2-14d41a5ab387`, worker `a8d2e919-c291-4e22-bb83-c09cdef72d7d`, auditor `2d9af7c0-fdec-4e0e-840d-dbe01738ce5b`.
- Cycle 2: researcher `c66a075b-1728-41cd-9128-1cb916c23778`, worker `d17794a7-2fde-4598-a3d1-7dedacfb5292`, auditor `a98248bb-6804-4716-b9d7-2c0513192441`.

Auditor decision (c2): COMPLETE. Sub-milestone `M-DAW-SPIKE-1/palette-instrument-determinism` closes at `validated/high` with per-instrument verdicts `surge_xt=STILL_GAP`; `dexed=STILL_GAP`; `sfizz=GREEN`. All 15 sufficiency criteria met.

Deliverables on disk.

- Code: `scripts/palette_probe/{surge_xt,dexed,sfizz,shared,run_all}.py` — 5 scripts, interpreter-guarded, no PRNG (AST-checked), no `sidcar_nonfactor` imports (AST-checked), no cycle-9 chain import (grep-verified).
- Data: `data/palette_probe/{instrument_determinism.tsv (3 rows, frozen labels), rubric_hash.txt, per_instrument/<inst>/{run1_wav_sha, run2_wav_sha, pinned_state.json, refinement.json, fetchability_ladder.jsonl}}` for each of the three instruments.
- Report: `docs/palette_instrument_determinism_report.md` (8 sections including per-instrument section, fetchability ladder, pinned-state format spec).
- Rubric: `docs/palette_instrument_determinism_rubric.md` (SHA-256 `75daa068aa804351db744cdb3a41df151ba682bbe` committed before any probe script landed).
- Tests: `tests/test_palette_instrument_determinism.py` (9/9 PASS, brief floor of 8 exceeded); `tests/test_integration_cross_branch.py` §45 (32 sub-checks, all PASS).

Load-bearing runtime evidence.

- Rubric SHA verified live: `75daa068aa804351db744cdb3a41df151ba682bbe3278c7c8cb8870a54ac7c96`.
- Per-instrument verdicts mechanically dispatched: `sfizz GREEN` (WAV SHA `4f9735d9...`, state SHA `1bda0de7...`, both equal × 2); `Surge XT + Dexed STILL_GAP` under the DawDreamer 0.9.0 `get_state()`=0-bytes root cause.
- Anchor preservation: cycle-9 chain untouched (grep-verified + §45 integration guard); cycle-13 batch pipeline untouched; prior batch anchors byte-identical.
- 9/9 branch tests + 32/32 §45 integration checks + all prior test suites unchanged.
- `promise_check` 0 ERRORS; `org_check` no new WARNs.

Ledger routing. Ten shadow-ledger events emitted at `/home/user/music-gen-instance/fork-cfc5009aca96/clone-0/promise_ledger.jsonl` in strict order — six named + two housekeeping + two bonus:

1. `cycle_31_launched (_run/cycle_31_launched_branch_A)`.

2. `_plan/register-palette-instrument-determinism` (bonus; registered rows on both plan tables).
3. `verdict_rubric_frozen_palette_determinism` (rubric SHA in narrative).
4. `palette_probe_scripts_landed`.
5. `palette_probe_run_complete` (per-instrument artefact list).
6. `M-DAW-SPIKE-1/palette-instrument-determinism verdict roll-up` (narrative: `surge_xt=STILL_GAP`; `dexed=STILL_GAP`; `sfizz=GREEN`).
7. `cycle_31_closed (_run/cycle_31_closed_branch_A)`.
8. `_archive/cycle-31-branch-A-scratch` (housekeeping).
9. `_infra/adopt-cycle31-tests` (housekeeping).
10. `_infra/anchor-guard-extended-cycle30` (opportunistic bonus — extended tests/fixtures/cycle28_util_shas.json with cycle_30_utilities; idempotent-if-same-content dedup guard applied).

All events use nested confidence: {level, rationale, assessor}, canonical narrative field, canonical run_id: run-2026-08-28T040704Z, UUID5 content-hash event_id auto-derived, two-arg `append_ledger_event(workspace, event)`. Auto-concat under the cycle-22 harness-namespacing fix; orphan-artefact WARNs on new artefacts clear at post-merge concat via the `_infra/adopt-*` mechanical pattern.

Standing anti-patterns unchanged (5). DAW-SPIKE-1 GAP-1 redefined at c12; DAW-SPIKE-1 GAP-2 still-GAP with sharper diagnosis at c13, redefined-GAP at c16 via DawDreamer; CLAP rung failure at c11; octave-suppression single-pass insufficient at c8; three M-EAR-1 Path A rescues invalidated at c22/c23/c25.

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at `/usr/lib/vst3/`; Dexed at `/usr/lib/vst3/`; sfizz_render CLI executable in PATH; basic-pitch 0.4.0 in `workspace/basic_pitch_venv/`. Single-thread BLAS pins throughout. Cycle-9 pinned DawDreamer chain untouched for 22 consecutive cycles.

Merge report. `/home/user/music-gen-instance/fork-cfc5009aca96/clone-0/merge_report.md` = 6460 bytes, byte-identical to `workspace/merge_report_cycle_31_branch_A.md` (verified via `os.stat` and content-hash).

Handoff. For the root conductor / next-cycle researcher (not for this clone): cycle 32 palette dispatch consumes sfizz via `sfizz_render_cli` as authoritative; routes Surge XT / Dexed voices through the cycle-13 fluidsynth GM pipeline as fallback. Sibling B's palette-assignment schema should accommodate the three `loader_pathway` enum values. Two follow-up peer sub-sub-milestones (`preset-refinement`, `dawdreamer-upgrade-probe`) queued but not auto-opened. Two housekeeping backlog items flagged upstream (`promise_check._parse_plan_milestones` full-cell match; anomalous `_run/report_cycles_32-34` ledger event). Standing constraints unchanged; α pinned; anti-patterns locked; egress still blocked; rated-audio unblock remains a straight-line consequence of the egress-ready state machine firing.

validated